00 — Platform Basis: Scope, Design Point, Shared Physics & Doctrines

v1.2 — foundation document for both tracks

Function: Everything both tracks share: deployment scope, the governing design point, the shared psychrometric argument, the generalized airflow–moisture model, the ventilation/CO_2 stack, the exhaust-vapor recovery doctrine, and the safety-critical requirements register. Track documents (10–12 liquid, 20–22 solid, 30 integration) reference this document rather than restating it.

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1. Scope — land and marine

The system targets **both land and marine deployment** in humid-tropical (and by extension any humid) climates. Design rules are written to the *harshest common case* so one document set covers both:

Aspect	Marine (governing case)	Land relaxations
Raw-water sink	Seawater ~29 °C (tropical surface)	Lake/river/well water, often cooler (22–28 °C tropics; lower elsewhere) → every sorbent floor improves (see §3 leverage). No water sink: evaporative fluid cooler (approaches wet bulb ~30 °C — near parity) or dry cooler (approaches dry bulb 32 °C — floors rise ~1.5–2 g/kg; state the penalty)
Motion	Heel/roll to ~15–20°; flooded-mode fallbacks, staged headers, anti-slosh rules	Stationary: motion provisions optional, retained as shipped hardware margin
Salt aerosol at intake	Mandatory intake filtration / sealed paths	Coastal: keep; inland: may relax intake filtering only
Chloride materials rules	Full two-worlds discipline (doc 11 §1)	**Never relaxed** — the desiccant itself is the chloride source
Envelope	~100 m³ occupied reference (yacht interior / small dwelling / cabin / shelter)	Same physics; volume thresholds per §6 ladder

Terminology: "platform" = vessel or land site; "raw-water sink" = seawater or its land equivalent. Marine values are used in all reference numbers; land is margin.

2. Governing design point (sole)

DP-A: 32 °C / 80% RH · ω = 24.2 g/kg · dew point 28.1 °C · raw-water sink ~29 °C. DP-A is the **continuous-duty maximum rating for every mode**. Milder ambients are operating margin (every duty term shrinks monotonically with ambient ω); rare excursions above DP-A are handled transiently — moisture-battery draw-down, setpoint relaxation, boost — never by sizing. A prior secondary point (30 °C / 75%) is retired; numbers derived from it in archived documents are historical records.

Saturation humidity ratios used throughout (Magnus, 101.325 kPa): 26 °C → 21.3, 27 °C → 22.6, 28 °C → 24.1, 29 °C → 25.6, 30 °C → 27.1 g/kg.

Air-density basis: volumetric-to-mass air conversions throughout the lineage use **ρ = 1.2 kg/m³** (standard air), not the ~1.15 kg/m³ of moist air at DP-A itself. Every mass flow — and every duty derived from one — therefore runs ~5% high. The direction is deliberate: it oversizes rather than undersizes. Stated here so it is an assumption on the record rather than one implied by the arithmetic.

3. The shared physics — why a desiccant is non-negotiable

Comfort in humid heat requires removing water from air, and only two things do that: a surface below the dew point (refrigeration) or a sorbent whose surface vapor pressure sits below ambient (a desiccant). At DP-A every ambient sink — raw water ~29 °C, outside air — sits **at or above the 28 °C dew point**, so:

- a sink-cooled condenser harvests ≈ zero water from ambient air;
- evaporative cooling (including the M-cycle) cannot go below the dew point — potent only once air is *pre-dried*;
- the desiccant breaks the floor because its surface vapor pressure is set by loading/concentration, not by a cold sink. The hard problem is **regeneration against near-saturated surroundings**, never capture.

Two sorbent families implement this: a $CaCl_2$ brine (liquid track, docs 10–12) and an AlFu MOF coated exchanger (solid track, docs 20–22). Their regeneration physics differ structurally — **finding X2**: the liquid track's sealed still has no purge stream, so its driving force (hot-pool Pv vs ~4–5 kPa sink-cooled condenser) stays positive at any pool temperature above ~40 °C and only *rate* degrades at low grade; the solid bed desorbs into a condensing purge (~25 g/kg) and hits a zero-driving-force wall below ~50 °C (F2). Consequence: **the liquid track is the solar-grade layer; the solid track is the high-duty waste-heat layer.**

Raw-water leverage (both tracks): cooler sink water deepens every floor — ~0.5–0.7 g/kg per °C on the brine side; land installs with cool groundwater get this for free.

4. The generalized airflow–moisture model

Closed steady-state balance replacing hand chains; the sorbent back-end enters only through the supply humidity ω_sup (AlFu ~8 g/kg; brine per the doc 12 floor matrix). Unknowns solved simultaneously:

1. $\omega\_cab = \omega\_sup + latent\_gains / S$ (cabin steady state)
2. $T\_sup = T\_in - \varepsilon\_dp \cdot (T\_in - T\_dp(\omega\_work))$ (M-cycle, $\varepsilon\_dp \approx 0.7$)
3. $S = Q\_sens / (c\_p \cdot (T\_cab - T\_sup))$
4. $Q\_wet = S \cdot c\_p \cdot (T\_in - T\_sup)$
5. $M = Q\_wet / \Delta h\_work$, $\Delta h\_work = h\_sat(T\_exh) - h(\text{working-air entry state})$
6. Sorbent duty = $(S - M)(\omega\_cab - \omega\_sup) + M(\omega\_amb - \omega\_sup)$

Load-bearing outputs at DP-A: solid track full-AC duty **~9–11 kg/h** (doc 20 §5); liquid mixed-mode duty **0.88 kg/h** (doc 12 §1); and **finding X1** — once-through ventilation and whole-cabin M-cycle cooling never compose (drying 370–750 m³/h of ambient supply costs 5.5–11 kg/h). A full transient model (task T2) supersedes this steady state; it does not replace it.

5. Ventilation & CO_2 — the four-layer stack (X6/X7/X9/X10, spec P17)

Specification (safety-critical): cabin CO_2 <1,000 ppm at all times, in every mode including sealed; alarm + forced boost at 2,000 ppm. Four occupants generate ~3.4 kg CO_2/day (**crew total**, not per person: 0.072 m³/h awake / 0.047 asleep); CO_2 leaves only with air exchanged, so recirculation, cascades, and ERVs move moisture, never CO_2.

- **Layer 0 — eliminate (X9): all-electric galley, no gas or combustion appliances aboard/on-site.** One 2 kW burner emits ~0.26 m³/h CO_2 (3.6× the crew) plus ~0.25 kg/h combustion water; induction deletes both (~2 kWh_e/day).
- **Layer 1 — ventilate efficiently.** 48 m³/h CO_2-governed fresh-air floor (12 m³/h·person; also carries bioeffluents, odors, O_2). **$\varepsilon\_lat \geq 0.8$ counterflow ERV** on the fresh/exhaust pair (CO_2 crossover <5%, test E); demand-controlled ventilation; **displacement ducting** — supply low to berths first, exhaust pickups high in every closable room (closed doors otherwise run +2,000–2,900 ppm over the main space); **per-room NDIR sensors, interlock keyed to the maximum reading**. Marginal cost of fresh air at DP-A: $(1-\varepsilon\_lat) \times 10.4$ g/kg ≈ 0.06 kWh_heat/day per m³/h at ε 0.8 (**ERV'd basis** — the bare-fresh equivalent is ~5× higher, and the two are quoted separately below).
- **Layer 2 — scrub the increment (X10): the CO_2 battery, required-PENDING test J.** Two-bed solid-amine thermal-swing sorbent (Lewatit VP OC 1065-class, or DIY PEI-on-silica) in the recirculation branch, post-dehumidification (45–55% RH is near-optimal for amine chemistry). Duty at the ventilation floor: ~1.6 kg/day; two ~3 kg beds on ~90-min half-cycles; ~91 m³/h at 50% single-pass capture rides the existing recirc flow (+50–150 Pa); **regeneration 85–95 °C off the same heat bus, ~1.0–1.3 kWh/kg → ~1.6–2.5 kWh/day** ; CO_2-rich purge vented out (§7 rule 1). Loaded beds store the overnight obligation (~0.3 kg) → regeneration is solar-window schedulable. **Bed chemistry follows the heat grade (X11):** the scrubbing duty is chemistry-agnostic; select the sorbent by the highest heat grade reliably available, top down. **≥ ~130 °C tap (waste-heat / reactor platforms): K_2CO_3-on-apolar-carbon TSA** — same bed envelope (~4–5 kg/bed at ~0.5–0.65 mmol/g working), carbonation *wants* the 45–55% RH placement (water is a reagent), regeneration 130–150 °C, food-grade chemistry with no amine-slip or oxidative-fade pathway (the breathing-air gate becomes an alkaline-carryover assay), required-PENDING test J-K; **alumina and MgO supports prohibited** (irreversible double-salt deactivation). **Solar grade only (60–95 °C): the amine resin above** — potash is equilibrium-dead below ~120 °C (an F2-analogue wall; the ETC's 93 °C ceiling cannot regenerate it). Combinations run the ladder for redundancy: a platform fitting potash as primary retains a small amine bed (or accepts the open-mode fallback) for sealed operation through a heat outage. **This is the only path that holds <1,000 ppm sealed.** Fallback if test J fails: oversized ERV + DCV at 120–190 m³/h → ~800–1,000 ppm, open conditions only, no sealed-mode guarantee. Rejected alternatives (quantified in doc 12 §5): zeolites (water outcompetes CO_2), seawater absorption (>6 m³/h equilibrated + re-humidifies the air), membranes at 0.1 kPa, algae (150+ kWh/day of light), liquid amines (slip into breathing air), chemical scrubbing (~15 kg/day soda lime — **emergency consumable only**).
- **Layer 3 — interlock.** CO_2-governed fresh damper with a **mechanical minimum stop** (no controller fault or economy setting can close ventilation below the floor); recirculation-only operation prohibited while occupied; scrubbing never substitutes for the ventilation floor.

Control law: optimize marginal ventilation against marginal scrubbing on heat availability — heat-rich, open the damper toward ~600–800 ppm; heat-scarce or sealed, the bed holds <1,000 at the floor.

CO_2 dose–response at the floor and below (4 occupants; why the interlock exists): 48 m³/h → 1,920 ppm awake; 36 → 2,420; 24 → 3,420; 12 → 6,420; sealed (infiltration only) → ~7,600. Each 12 m³/h of fresh air cut saves ~3.8 kWh/day of **bare-fresh** heat at DP-A (≈0.32 kWh/day per m³/h un-recovered; with the ε 0.8 ERV fitted, the same cut saves only ~0.7 kWh/day) — a standing economic temptation the interlock forecloses.

6. Volume-threshold ladder at DP-A (liquid track; solid track is full-AC class)

Mode	Serviceable volume	Limit type
Once-through cascade M-cycle cooling	~2–5 m³ per outlet (74–112 W)	dew point of working air
Unattended, 1 contactor, holding 13 g/kg	~90 m³ (×2 per contactor)	contactor capacity vs 0.1 ACH
Unattended, 1 contactor, mold-safe (60–65% RH)	~160–215 m³	"
Once-through occupied (pressure integrity)	~110–240 m³	supply ≥ 2–3× infiltration
Mixed-mode occupied (baseline)	**~300–500 m³** with a 3-cell bank	bank capacity + heat budget
Whole-cabin AC (recirc + hot-regen brine)	100 m³-class at ~10.6 kg/h, ~11 kW heat	deferred — converges with the solid track

Diagram: `diagrams/dpa-volume-thresholds.svg`.

7. Exhaust-vapor recovery doctrine (X8)

All *deliberately saturated* process exhausts terminate in distillation recovery when marginal regeneration heat is available; heat is recaptured where practical. Diagram: `diagrams/exhaust-recovery-doctrine.svg`.

1. **No double duty.** A stream is a rejection sink *or* a recovery source, never both — recapturing deliberately exported moisture re-imports the load just paid to reject.
2. **ERV protection.** The ducted room exhaust is reserved as the ERV moisture sink; the ERV outlet leaves near-saturated *by design* and is discharged un-recaptured.

3. **Saturated process exhausts end in recovery:** the M-cycle working stream closes on the desiccant loop (solid track: design intent, doc 20 §8; liquid track: valved free-heat option at ~1.3 kWh/L, doc 10 §6); the air-swept regeneration exhaust (degraded mode) gains a raw-water condenser — its condensate is **technical-grade only, PENDING TDS assay** (entrainment).
4. **Heat-recapture order:** DHW tank first (demand-capped ~1–3 kWh/day) → bed-to-bed / brine-to-brine recuperation → raw-water sink last. Adsorption heat releases only a few kelvin above the sink — reject it, don't chase it.

8. Safety-critical requirements register

Any build omitting these departs from this design:

1. CO_2 interlock per §5: <1,000 ppm target, 2,000 alarm/boost, per-room max sensing, mechanical minimum stop on the fresh damper.
2. No gas or combustion appliances in the conditioned envelope (X9).
3. Aerosol control chain + steel-coupon acceptance (test B) before any liquid contactor connects to breathing air (doc 11 §3).
4. CO_2-sorbent breathing-air assay before any bed connects to breathing air: amine/ammonia slip (test J) for amine beds; alkaline particulate/mist carryover (test J-K) for carbonate beds.
5. Distillate potability: independent water-quality test before regular consumption; air-swept-condenser water never enters potable or M-cycle-feed service before a TDS assay clears it.
6. Crystallization interlock: 42 wt% high-SG cutoff unless brine ≥22 °C guaranteed; 43 wt% storage cap (doc 11 §4).
7. Two-worlds materials rule on every brine/raw-water-wetted component (doc 11 §1).

9. Conventions

Confidence grades on every quantitative claim: **measured** · **procurement-grade** · **sizing-grade** · **PENDING <test/task>**. DP-A on every number. Diagrams: Mermaid inside documents; the SVG architecture set lives in `diagrams/`. Psychrometric chains as state tables. Surgical edits with version bumps.

Version history

- **v1.0** — New lineage. Consolidates the shared basis from the archived core-01 §1, 01 §1–2, and 06 §§1–3, 9–11; land+marine scope made explicit; DP-A sole; CO_2 stack (P17, X9/X10), X8 doctrine, and safety register carried in place.
- **v1.1** — X11: heat-grade-dependent CO_2-bed ladder added to §5 Layer 2 (K_2CO_3/apolar-carbon TSA for ≥ ~130 °C taps, test J-K; alumina/MgO support prohibition; amine bed retained at solar grade and as outage fallback); safety-register item 4 generalized to cover the carbonate alkaline-carryover assay.
- **v1.2** — Clarifying pass from the parameter register (doc 50 §3.5), no design figure changed: the $\rho = 1.2$ kg/m³ air-density basis stated explicitly in §2; the CO_2 generation rate in §5 labelled *crew total* to foreclose a 4× misreading; the §5 marginal-ventilation costs labelled ERV'd vs bare-fresh, which are on different bases and were previously quoted side by side without distinction.